

Solar Cookers for All (Sc4all)

*Een samenwerking tussen UHasselt &
Universiteit van Lubumbashi (DRC)*

Dag van de Wetenschap

UHasselt

23 november 2025



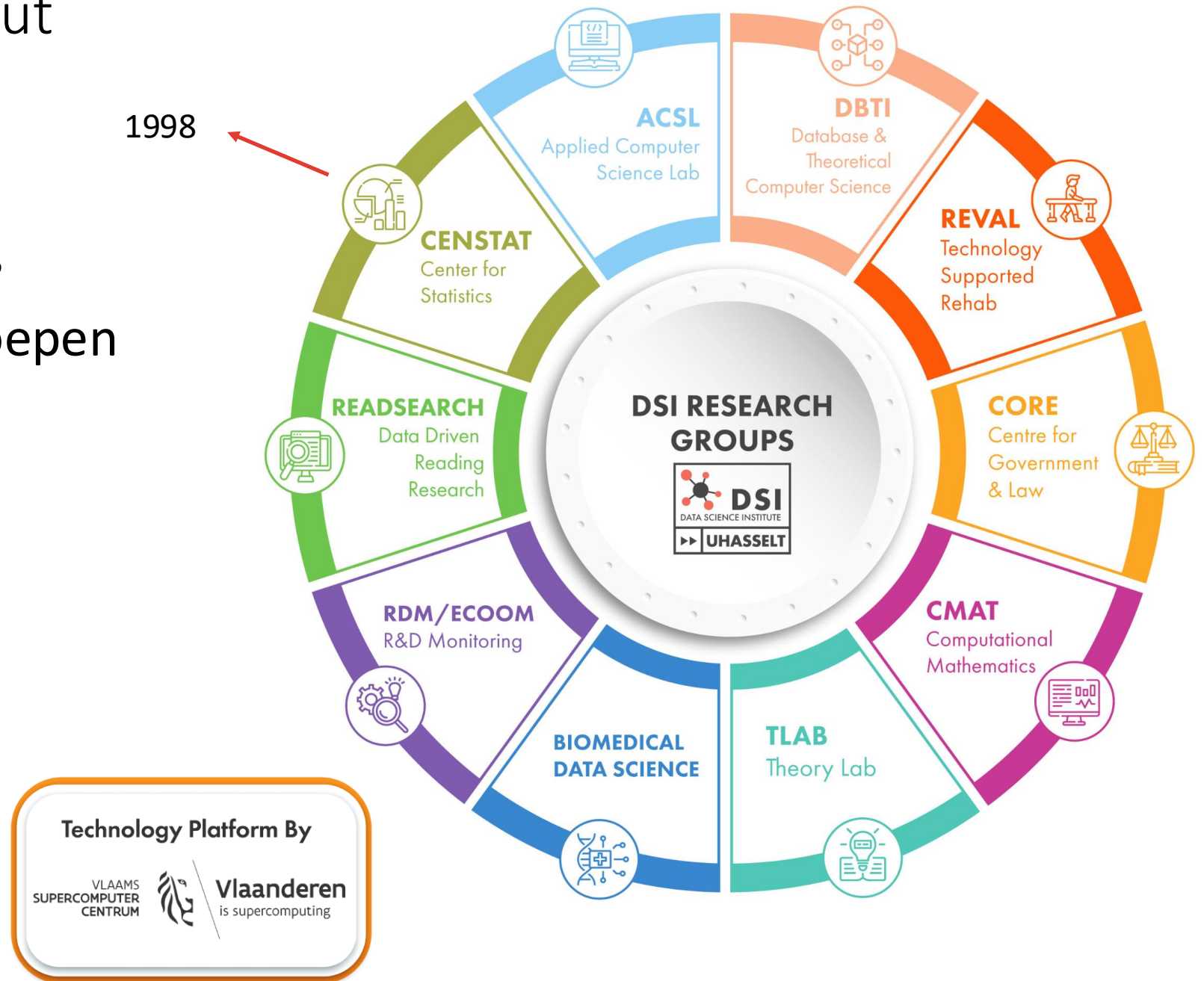
WWW.UHASSELT.BE/DSI

<https://www.uhasselt.be/en/instituten-en/dsi/solar-cookers-for-all>

Data Science Instituut

- 150 onderzoekers
- 10 onderzoeksgroepen
- 5 faculteiten

Opgericht in 2019





Zonnekokers ... wat?



Zonnekokers ... waarom?



Sustainable Development Goals (SDGs)

United Nations 2015-2030



Geen armoede

Geen honger

Goede gezondheid

Onderwijskwaliteit

Gender-gelijkheid

Water

Betaalbare propere energie

Economische groei

Industrie & innovatie

Reductie in ongelijkheden



Duurzame steden

Redelijke consumptie

Klimaatactie

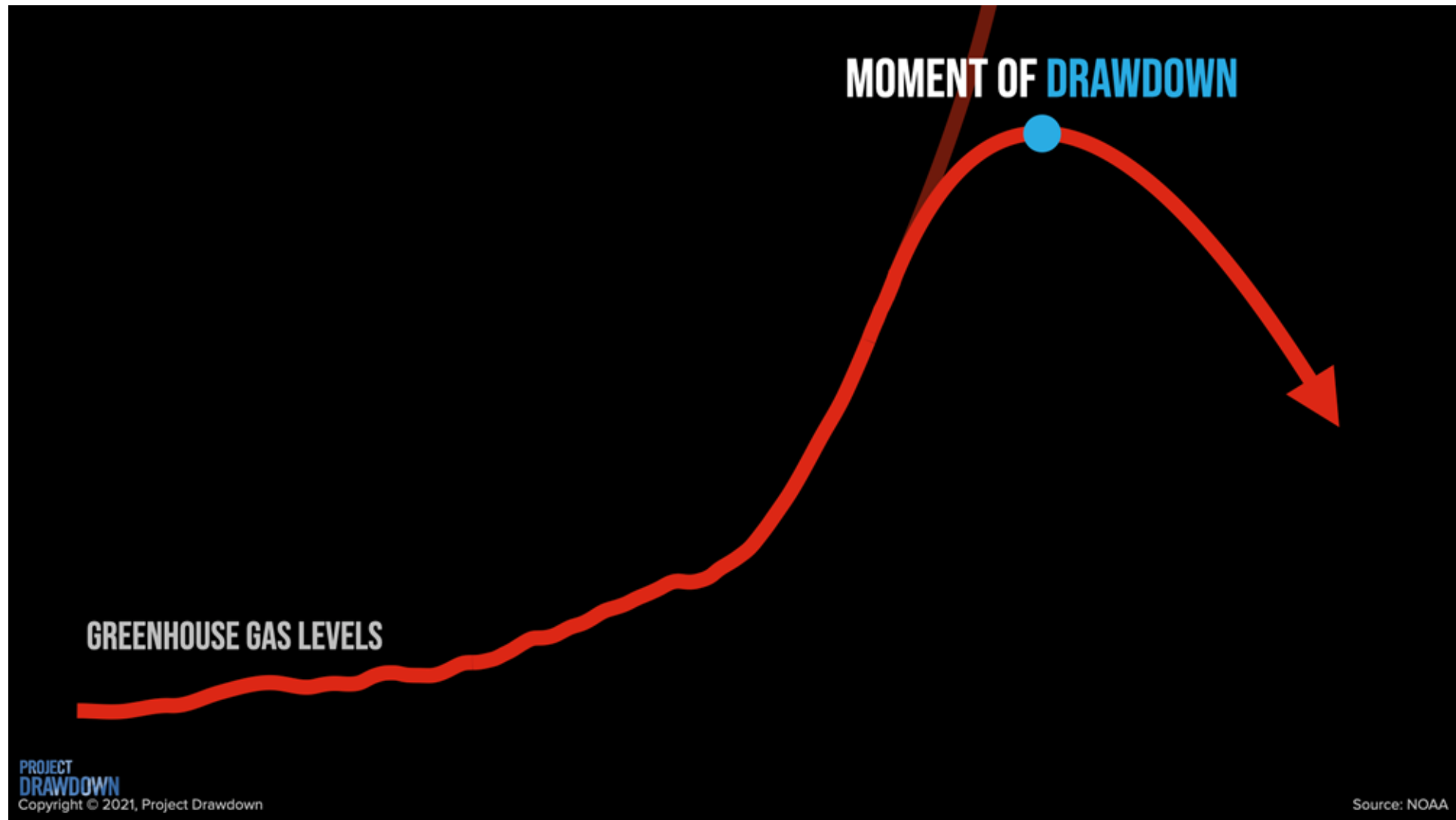
Water-leven

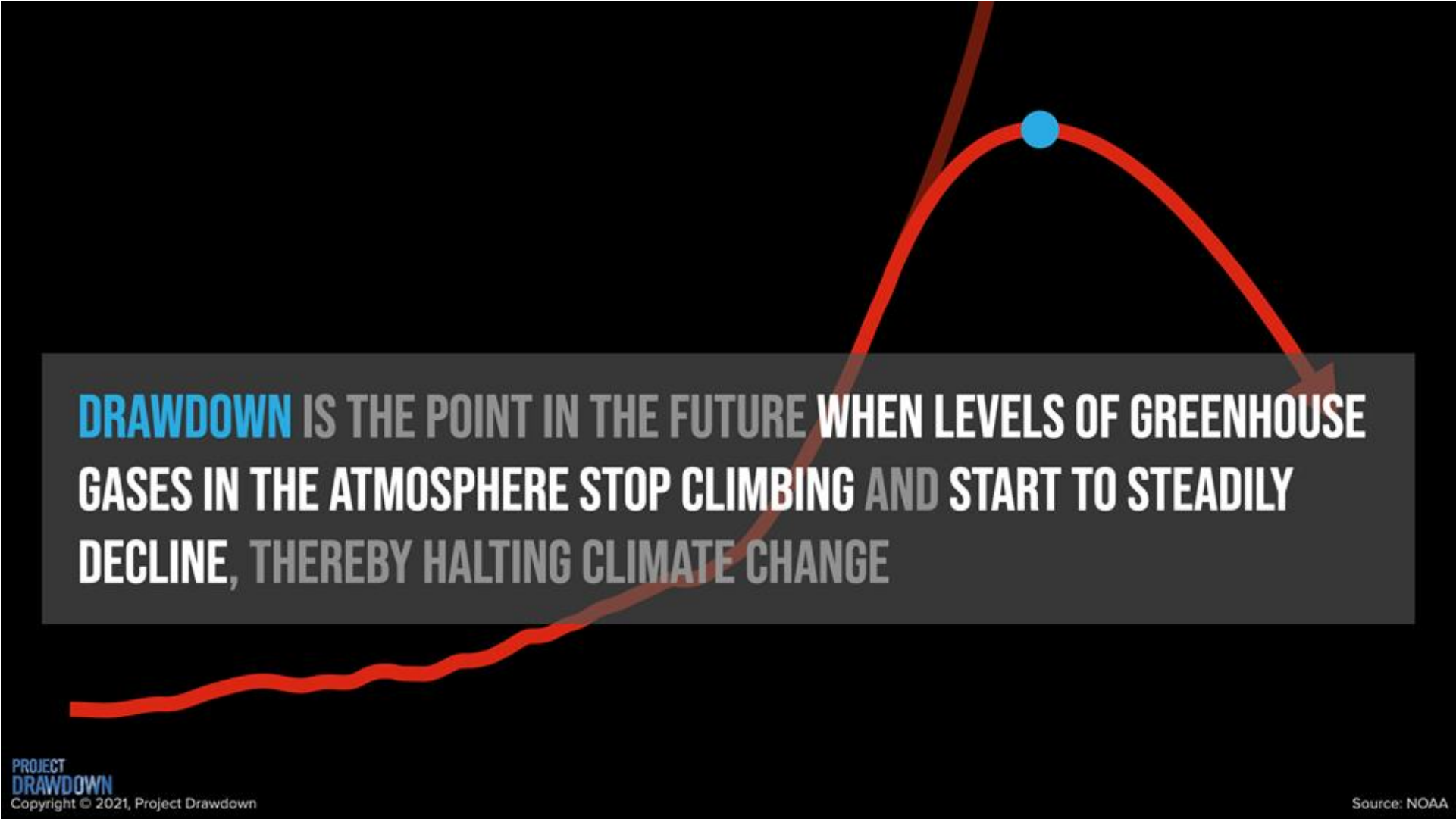
Land-leven

Vrede

Partnerships voor SDGs



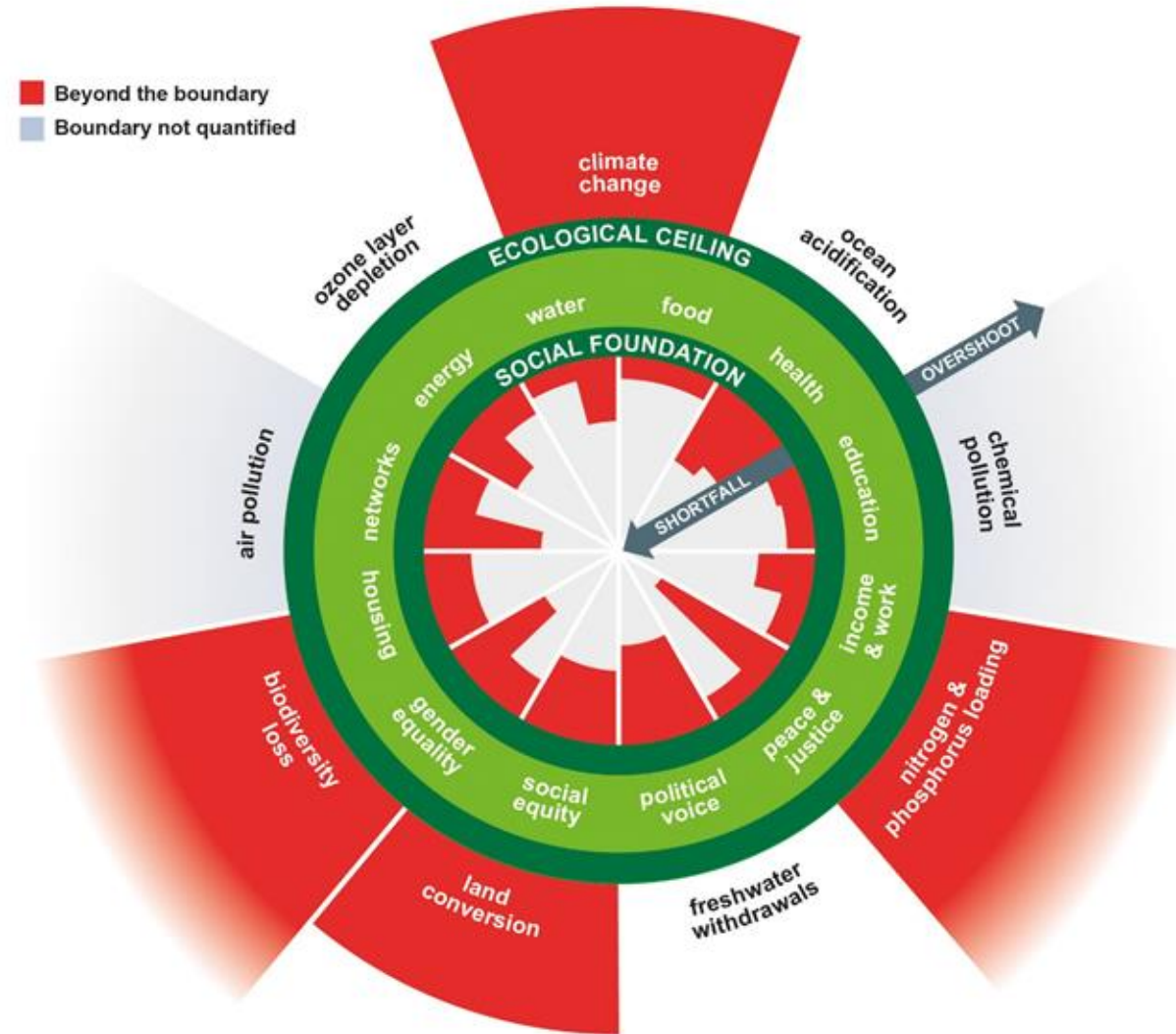




DRAWDOWN IS THE POINT IN THE FUTURE WHEN LEVELS OF GREENHOUSE GASES IN THE ATMOSPHERE STOP CLIMBING AND START TO STEADILY DECLINE, THEREBY HALTING CLIMATE CHANGE

Donut Economy (Selfie of the World)

Een circulaire economie is cruciaal



Kate Raworth economist at Environment Change Institute Oxford

2015 UN Climate Change Conferentie Parijs, Frankrijk



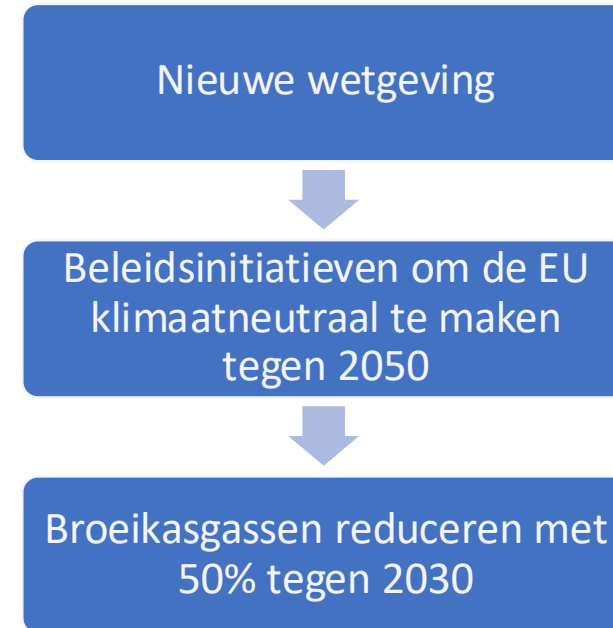
Parijse klimaatakkoorden

- Internationaal verdrag rond klimaatverandering
- Aangenomen in 2015 door 196 landen wereldwijd

Doelstellingen

- Beperking opwarming tot 1,5°C
- Vermindering broeikasgassen
- Adaptatie
- Nationale klimaatplannen
- Financiële steun

Europese Green Deal - Klimaatneutraliteit



“Europe’s man on the moon moment”

Ursula von der Leyen, Voorzitter van de Europese Commissie

Tegenhanger van de Green Deal in Amerika

“WASHINGTON — The Senate passed legislation on Sunday that would make the most significant federal investment in history to counter climate change and lower the cost of prescription drugs,...”. Emily Cochrane, New York Times 7 Aug 2022.

Inflation reduction act



- \$369 biljoen gependend gedurende een decennium
- Doel: 30% reductie in broeikasgassen
- Echter, geen 50% reductie tegen 2030 zoals beloofd in Parijs
- 20% extra reductie moet door industrie worden bekomen

Huidige situatie in België en daarbuiten

- **Stijgende wereldwijde uitstoot:** wereldwijde CO₂-uitstoot door fossiele brandstoffen naar verwachting met 1,1% stijgen in 2025
- Huidige uitstoot bereikt **recordhoogte van 38,1 miljard ton CO₂**
- **Parijsakkoord:** Om de opwarming te beperken tot 1,5°C moeten de wereldwijde emissies uiterlijk vóór 2025 hun piek bereiken en in 2030 met 43% dalen
- Huidige trajecten liggen **niet** in lijn met deze klimaatdoelen
- **Grootste uitstoters:** China blijft het land met de grootste uitstoot, hoewel er tekenen zijn van een mogelijke afvlakking van de groei
- In de VS nam de uitstoot toe, terwijl deze in België en in de EU daalde

2025 UN Climate Change Conferentie Belém, Brazilië

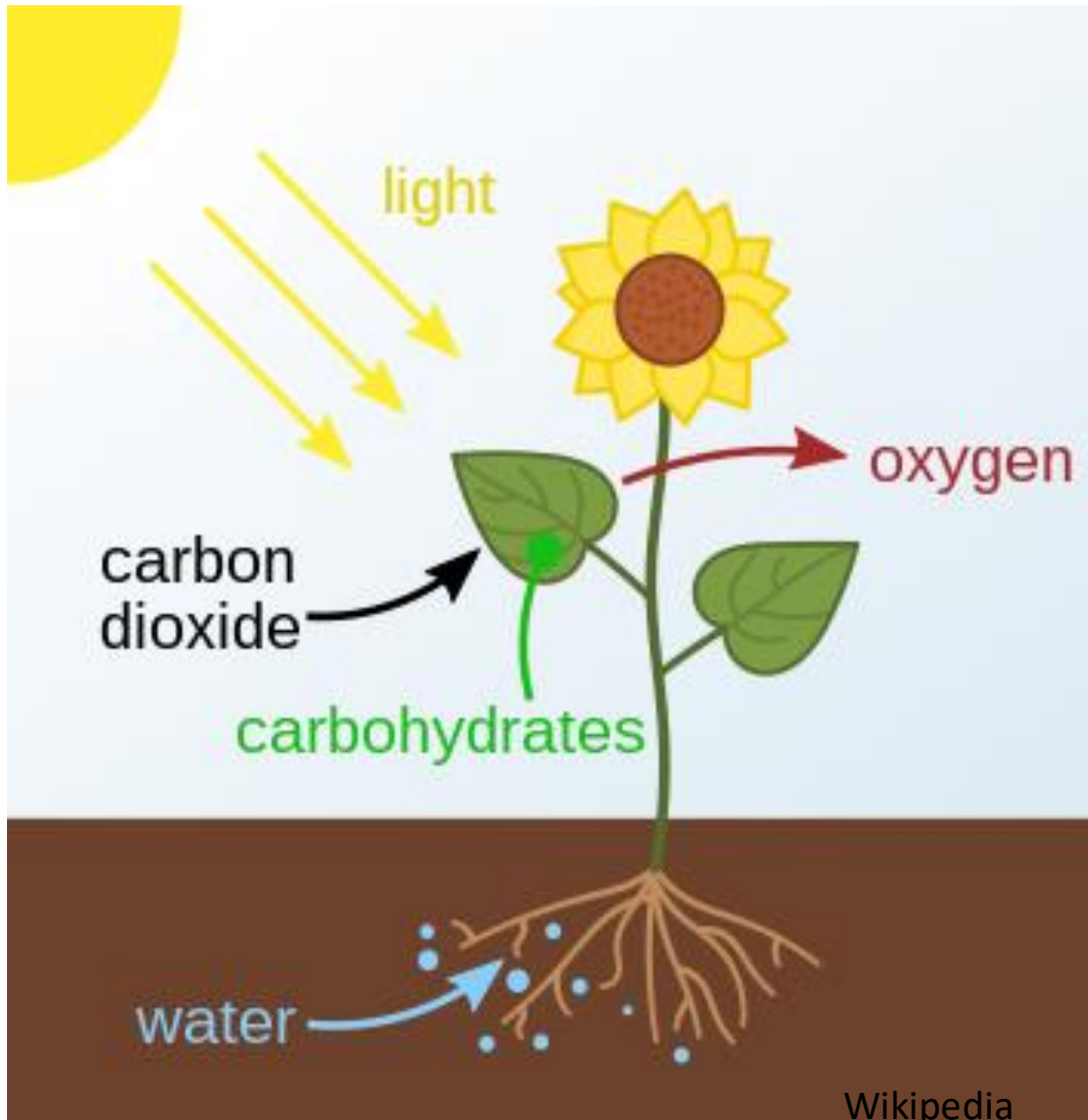


LIVE Teleurstelling én opluchting na "zwak"
klimaatakkoord: "Onvolmaakt, maar
noodzakelijk"

- Het slotakkoord belooft **3 keer meer geld voor kwetsbare landen** om zich te wapenen tegen klimaatverstoring.
- Fossiele brandstoffen zijn de grote afwezige in de slotverklaring → geen stappenplan om af te raken van olie, gas en steenkool

Geen sterke taal rond **strijd tegen ontbossing**

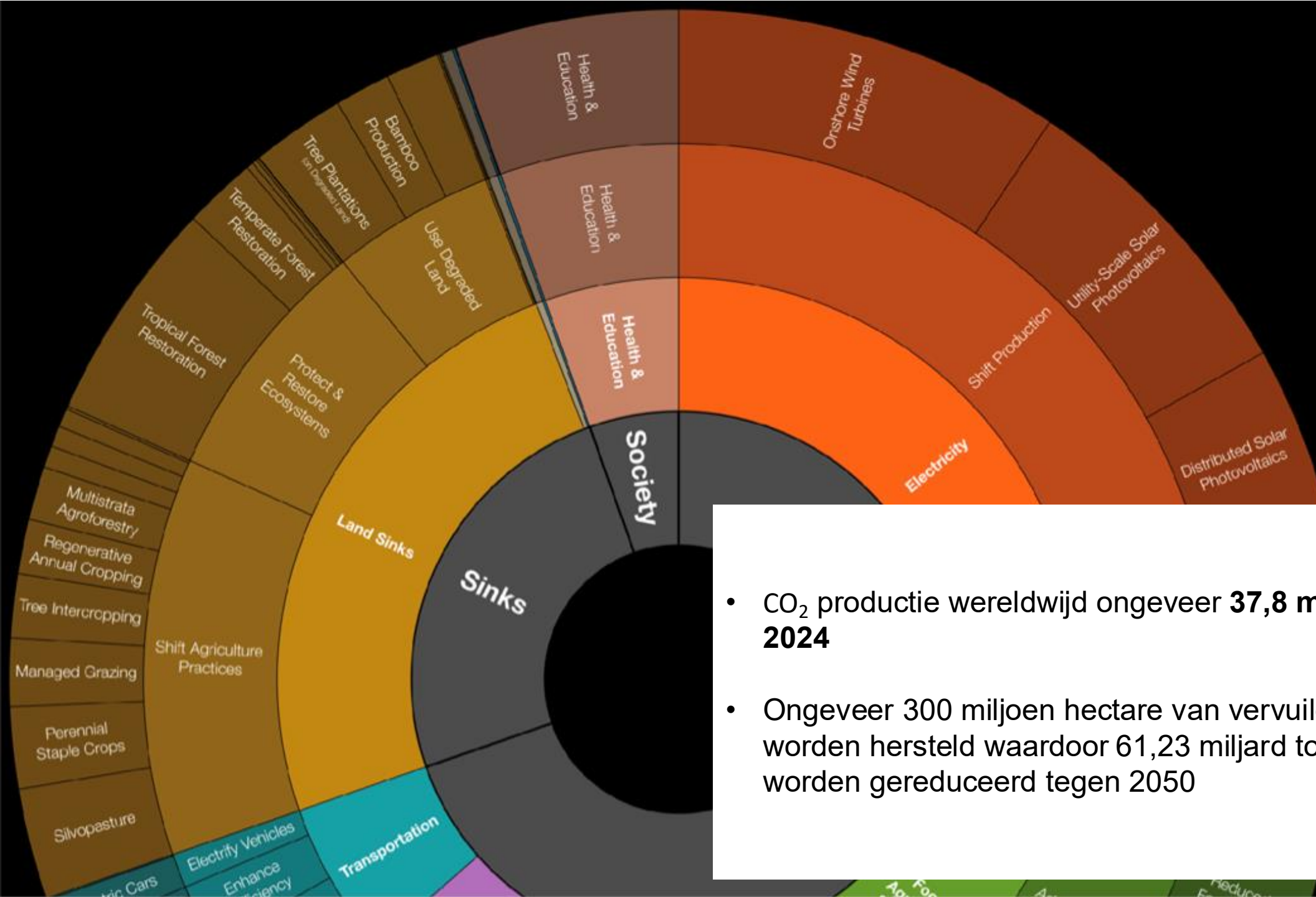




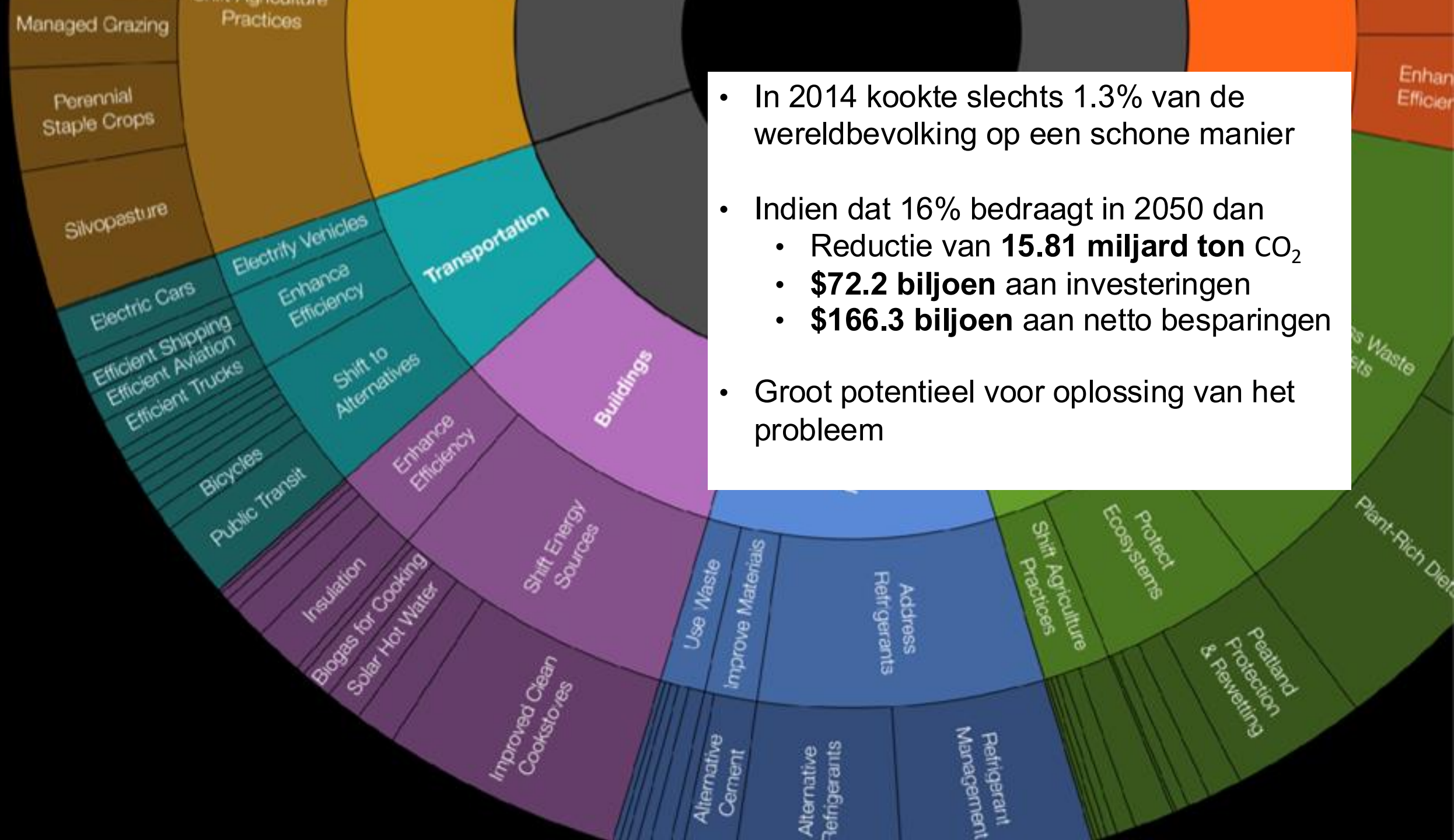
Fotosynthese

Proces waarbij planten **zonlicht gebruiken** om CO_2 en water om te zetten in zuurstof

- 0.9 ton CO_2 per m^3 bos opgeslagen
- 1 boom slaat 25 kg CO_2 op per jaar
- 1 hectare tropisch woud slaat 440 ton CO_2 op



- CO₂ productie wereldwijd ongeveer **37,8 miljard ton in 2024**
- Ongeveer 300 miljoen hectare van vervuilde grond kan worden hersteld waardoor 61,23 miljard ton aan CO₂ kan worden gereduceerd tegen 2050



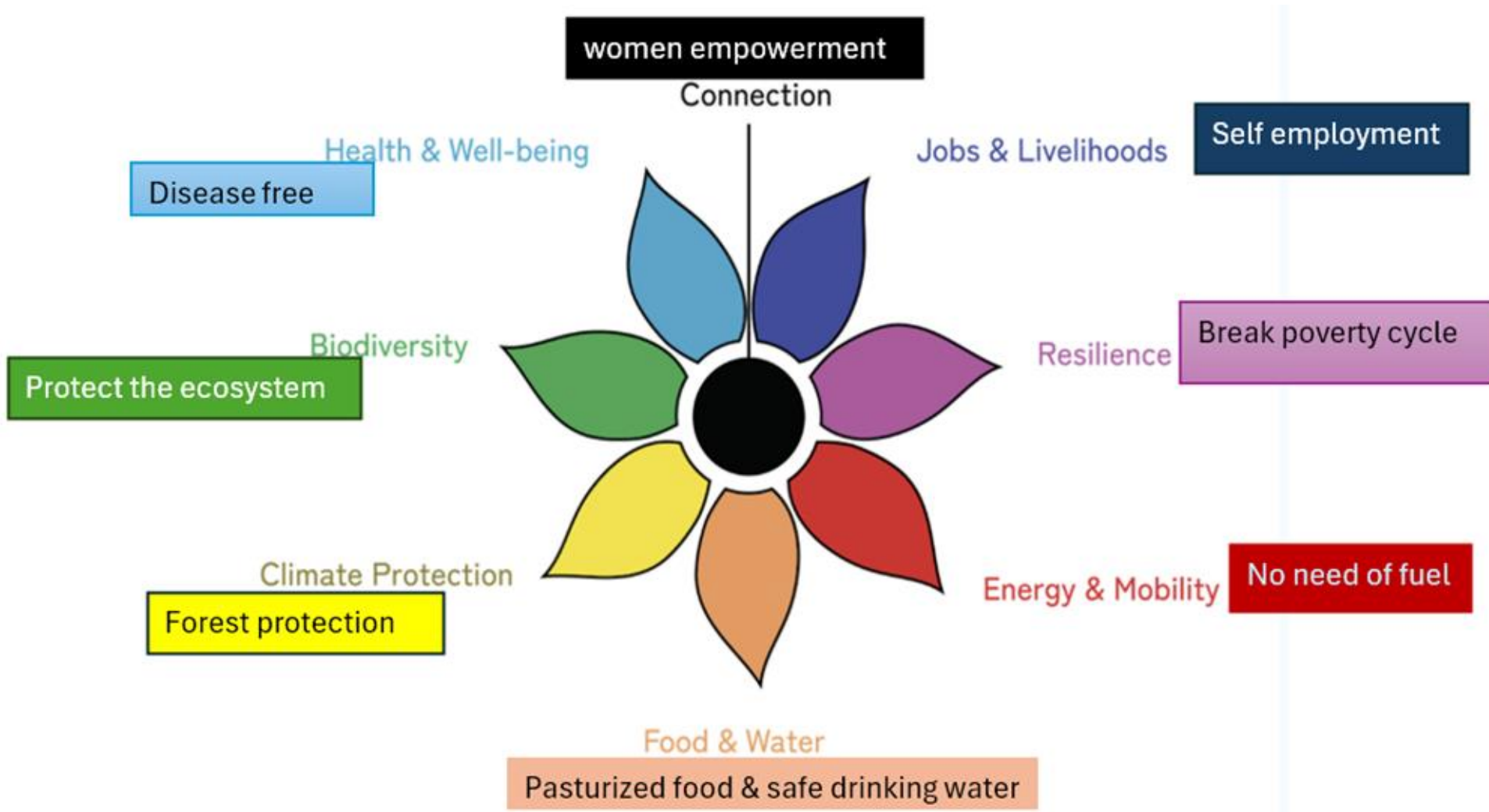
- In 2014 kookte slechts 1.3% van de wereldbevolking op een schone manier
- Indien dat 16% bedraagt in 2050 dan
 - Reductie van **15.81 miljard ton** CO₂
 - **\$72.2 biljoen** aan investeringen
 - **\$166.3 biljoen** aan netto besparingen
- Groot potentieel voor oplossing van het probleem

Zonnekokers

- Alternatief voor het gebruik van houtskool voor thuisgebruik
- Gezond en veilig in gebruik
- Geen negatieve impact op het klimaat
- Goede performantie nodig
- (Relatief) goedkoop
- Ruw materiaal van lokale gemeenschap
- Hoofdzakelijk gerecycleerd materiaal
- Lokaal gemaakt



Oplossingen: Adoption of solar cooker FLOWER*



Elizabeth sawin (<https://www.multisolving.org>)

Solar Cookers for All (Sc4all) project



Sc4all project

1. Ontwikkeling van zonnekoker prototypes (R&D)
2. Dataverzameling en performantie testen
3. Statistische innovatie
4. Workshops en studentenmobiliteit

Solar Cookers for All (Sc4all) project



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Solar Cookers for All

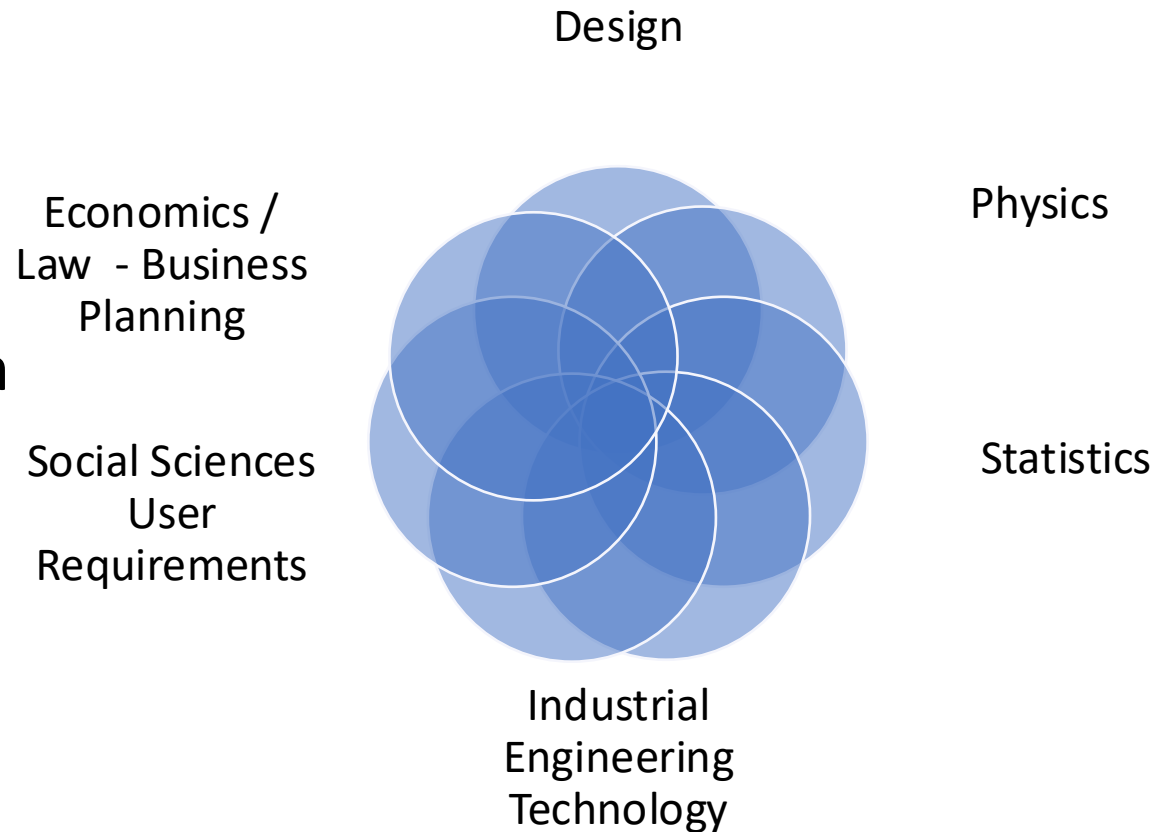
Learn more about SC4All



Solar Cookers for All (Sc4all) project

internationaal en multi-disciplinair team

- Economie
 - Rechten
 - Sociale wetenschappen
 - Fysica
 - Statistiek en datawetenschappen
 - Electronica
 - Ontwerpwetenschappen
-
- **Samenwerking tussen UHasselt en Universiteit van Lubumbashi (UniLu, DRC)**



Ontwikkeling van zonnekoker prototypes

- Bouwen van **prototypes**
- Uitvoeren van **experimenten**
- Creeëren van **instructies** voor het toekomstig lokaal construeren van zonnekokers



Ontwikkeling van zonnekoker prototypes

- Materiaalonderzoek:
 - Vertrekken van materialen die **lokaal beschikbaar** zijn
 - Bijv. blikjes



Ontwikkeling van zonnekoker prototypes

- Constructie van prototypes @UHasselt en UniLu



Dataverzameling en performantie testen

Performance Evaluation Process (PEP)*



ASAE S580.1 NOV2013
Testing and Reporting Solar Cooker Performance



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Dataverzameling en performantie testen



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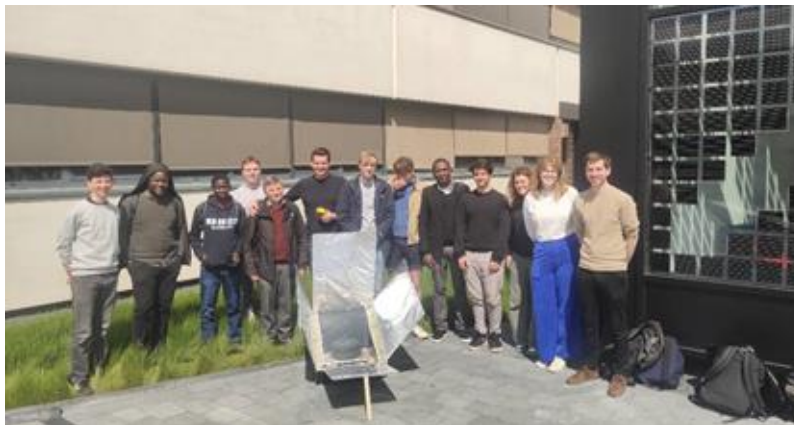
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Dataverzameling en performantie testen





Testen van de commerciële *Fornelia solar oven* voor de bepaling van een referentie qua prestatie (42 Watts)

Commerciële zonnekokers



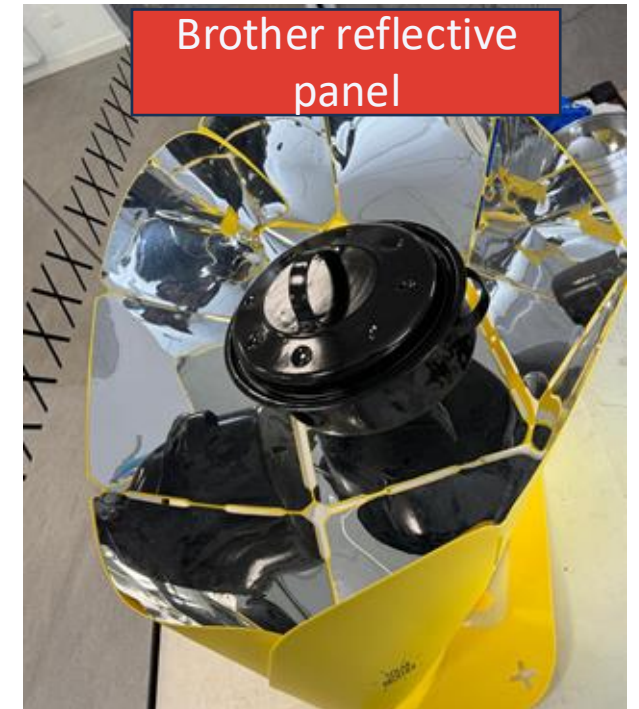
- Kooktijd 10 minuten
- 169.54 Watt

Data Science Institute



- Kooktijd 30 minuten
- 41.6 Watt

www.UHasselt.be/DSI



- Kooktijd 50 minuten
- 15.35 Watt

Parabolische zonnekokers

Prototype 2/3



- Kooktijd: 30 minute
- 10.10 Watt



Prototype 4

- Kooktijd: 30 minute
- 11.35 Watt

Oven prototypes

Oven Prototype 5



- ❖ Time to cook n/a
- ❖ 33.71 Watt

Oven Prototype 1



- ❖ Time to cook n/a
- ❖ -11.60 Watt

Oven Prototype 2



- ❖ Kooktijd: 90 minuten
- ❖ 3.24 Watt

Hybride prototypes



Statistische innovatie

Future proef award voor twee master thesissen in **Statistics and Data Science** rond nieuwe technieken voor de analyse van experimentele gegevens

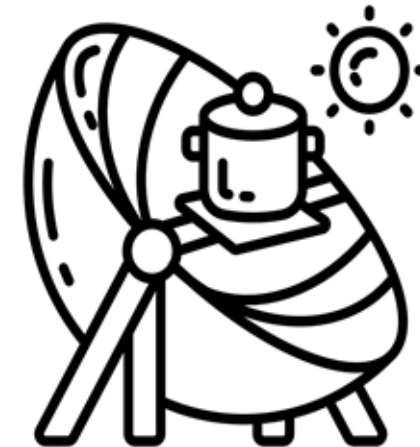
- *Abdulaziz Mwandu*: Design and statistical analysis of experiments for determining the standardized performance of solar cooker appliances
- *Felista Cosmas Kauki*: The use of survival analysis for the determination of the performance of solar cookers in Africa

Methodologische aanpak

Standard method (Simple linear regression)

Extension of standard method (Linear mixed models)

Novel statistical method borrowed from survival analysis (Cox regression)



Felista Cosmas Kauki & Abdulaziz Mwandu - UHASSELT

THE MASTERPROOF

"Design and statistical analysis of experiments for determining the standardized performance of solar cookers in Africa"



Statistische innovatie

Andere masterproeven in Statistics and Data Science

- *Waweru Mwaura Patrick*: Sample size determination of experiments for the determination of standardized performances of solar cooking appliances
- *Rashidi Baraka*: Innovative statistical analyses methods to quantify the performance of novel solar cooker appliances produced in resource-limited settings: A comparative approach



Descriptive statistics

Test dates and experimental time intervals

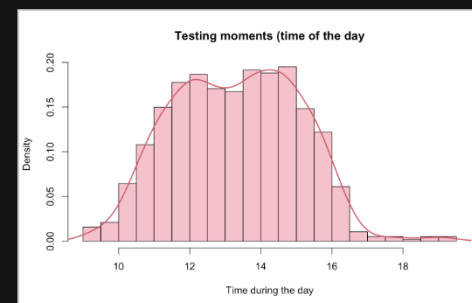
In total, data on 27 test dates were available. During these test dates, the majority of the tests were performed before 17h00.

[Hide](#)

```
time_stamps = paste(gsub("\\.", "-", solar_cooker_data$TestDate), solar_cooker_da
datetime_info = as.POSIXct(time_stamps)
time_info = strptime(datetime_info, format='%H:%M:%S')

dtTime <- as.numeric(datetime_info - trunc(datetime_info, "days"))
solar_cooker_data$dtTime = dtTime

hist(dtTime, xlab = "Time during the day", col = "pink",
      breaks = 20, freq = FALSE, main = "Testing moments (time of the day)")
lines(density(dtTime), col = 2, lwd = 2)
```



Workshops



R training at UniLu



- Het Sc4all team geeft **training aan UniLu studenten** over hoe de prestatie van nieuwe prototypes kan worden geëvalueerd en vergeleken met bestaande zonnepanelen



Kennisoverdracht en betrokkenheid van de gemeenschap

- **Bewustmakingssessies** in Lubumbashi
- Methoden en innovaties gepresenteerd aan vertegenwoordigers van de REFED-groep en **deelnemers uit de lokale gemeenschap**





Workshop in Altötting (Duitsland)

- Constructie van een SK14 zonnekoker en studie naar Zeolite koelkast

Trip naar & leven in Lubumbashi

Internship experience!



Studentenmobiliteit



Studentenmobiliteit

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Juli 2024





Stage in Tanzania



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In de toekomst

- Installatie van een **werkplaats in UniLu**
- Onderzoeken vanuit het **perspectief van de consument**
- Onderzoek naar:
 - Goedkope coatings
 - Koeling
 - Prototype voor fornuizen
 - ...

Stay tuned!

Stage in Tanzania







Université de Lubumbashi

